Raw decoder excerpt (continued) *preprocessing/pingstation_raw_decoder.py*

```
        raw = int.from_bytes(frame_bytes[idx:idx + 3], "big")
        idx += 3
        sign = -1 if (raw & 0x800000) else 1
        value = raw & 0x7FFFFF
        return sign * value, idx

    def _decode_compressed_vrs(self, frame_bytes):
        """
        Decode one Compressed VRS message.
        Returns dict of decoded fields or None if frame doesn't match.
        """
        if len(frame_bytes) < 9:
            return None
        if frame_bytes[0] != len(frame_bytes):
            return None

        icao = frame_bytes[4:7].hex().upper()
        flags = int.from_bytes(frame_bytes[7:9], "little")
        idx = 9
        out = {"icaoAddress": icao}

        if flags & self._F_CALLSIGN:
            if idx >= len(frame_bytes):
                return None
            n = frame_bytes[idx]
            idx += 1
            if idx + n > len(frame_bytes):
                return None
            callsign = frame_bytes[idx:idx + n].decode("ascii", errors="ignore").strip().upper()
            callsign = re.sub(r"[^A-Z0-9]", "", callsign)
            idx += n
            if callsign:
                out["callsign"] = callsign

        if flags & self._F_ALTITUDE:
            altitude_ft, idx = self._read_float_int24(frame_bytes, idx)
            if altitude_ft is None:
                return None
            out["altitudeMM"] = int(round(float(altitude_ft) * 304.8))

        if flags & self._F_GROUNDSPEED:
            if idx + 2 > len(frame_bytes):
                return None
            groundspeed_knots = int.from_bytes(frame_bytes[idx:idx + 2], "little", signed=True)
            idx += 2
            if groundspeed_knots >= 0:
                out["horVelocityCMS"] = int(round(groundspeed_knots * 0.514444 * 100))

        if flags & self._F_TRACK:
            if idx + 2 > len(frame_bytes):
                return None
            track_tenths = int.from_bytes(frame_bytes[idx:idx + 2], "little", signed=True)
            idx += 2
            if track_tenths >= 0:
                heading_deg = (track_tenths / 10.0) % 360.0
                out["headingDE2"] = int(round(heading_deg * 100))

        if flags & self._F_LAT:
            if idx + 4 > len(frame_bytes):
                return None
            out["latDD"] = struct.unpack("<f", frame_bytes[idx:idx + 4])[0]
            idx += 4

        if flags & self._F_LON:
            if idx + 4 > len(frame_bytes):
                return None
            out["lonDD"] = struct.unpack("<f", frame_bytes[idx:idx + 4])[0]
            idx += 4

        if flags & self._F_VSPEED:
            if idx + 2 > len(frame_bytes):
                return None
            vs_fpm = int.from_bytes(frame_bytes[idx:idx + 2], "little", signed=True)
            idx += 2
```